

```
[1,] 9 8 7
[2,] 6 5 4
[3,] 3 2 1
```

To multiply matrices M and N and store the result in matrix MM, the following commands can be used:

```
> MM <- M %*% N
> MM
      [,1] [,2] [,3]
[1,] 216 234 252
[2,] 177 192 207
[3,] 138 150 162
```

 **Note:** The system won't crash if the matrices are not conformable for multiplication.

If the commands typed are...

```
> SM <- M * N
> SM
```

...then the output will be:

	[,1]	[,2]	[,3]
[1,]	90	66	36
[2,]	104	70	30
[3,]	112	68	18

→

SM <- M * N	3 x 11	3 x 12	
	9 x 10	6 x 11	
	8 x 13	5 x 14	2 x 15
	7 x 16	4 x 17	1 x 18

Figure 2: Output of M * N

To find the determinant of M and display the result, use the following commands:

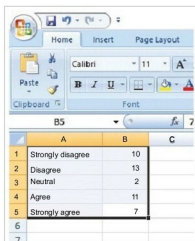
```
> d <- det(M)
> d
```

To find the inverse of matrix M, use the following command:

```
> MI <- solve(M)
```

To find the rank of matrix M, type in the following command:

```
rm <- qr(M)
> rm
```



	A	B	C
1	Strongly disagree	10	
2	Disagree	13	
3	Neutral	2	
4	Agree	11	
5	Strongly agree	7	
6			
7			

Figure 3: Sample data in the csv file

A typical output is shown below:

```
$qr
      [,1]      [,2]      [,3]
[1,] -13.9283883 -0.7590895 -3.589791e+00
[2,]  0.5743665  0.5275893  1.055179e+00
[3,]  0.5025707  0.9589395  1.280443e-16
```

```
$rank
[1] 2
$graux
[1] 1.646162e+00 1.283611e+00 1.280443e-16
$pivot
[1] 1 2 3
attr(,"class")
[1] "qr"
```

For combining matrices M and N (columnwise), use the following command:

```
> cbind(M,N)
```

The output will be as follows:

```
      [,1] [,2] [,3] [,4] [,5] [,6]
[1,]  9  6  3 10 11 12
[2,]  8  5  2 13 14 15
[3,]  7  4  1 16 17 18
```

For combining matrices M and N (row wise), use the following command:

```
> rbind(M,N).
```

 **Note:** 1) To get a list of all the variables that have been defined in a R session, use the `ls()` command.
2) To display the warnings being generated by a code snippet, use `warnings()`.
3) To exit from R, use `q()`.

Pie charts in R

Assume that a csv file (C:\Users\faculty\Documents\info.csv) contains the data given in Figure 3.

A pie chart can be drawn based on the data in Figure 3, with the following commands:

```
> d<- read.csv("C:/Users/faculty/Documents/info.
csv",header=FALSE,sep=",")
# for assigning the data in the file info.csv to d
> d # for displaying the value of d
```

R automatically assigns names to the rows and columns. The row names and column names can be displayed using the following commands:

```
> rownames(d)
[sample output] "1" "2" "3" "4" "5"
> colnames(d)
[sample output] "V1" "V2"
```

To display the values of V1 and V2, the following commands can be used: